Optimal Retinal Ganglion Cell Allocation under Fixed Channel Budgets

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Introduction

The mammalian retina must compress a high-dimensional visual scene into a limited number of retinal ganglion cell (RGC) spike trains along the optic nerve. Given the optical constraints of the eye lens, such as aberrations, input image statistics contain varying local spatial complexity around the fovea. Furthermore, the primate fovea dedicates 80% of ganglion cells to the central visual field (Kolb, 2012). We hypothesize that a nonuniform allocation is optimal from an information-theoretic perspective. We model RGC distributions across retinal regions as a fixed-budget resource allocation problem and compare allocation strategies against rate-distortion theory (Ganguli & Simoncelli, 2010, 2014).

Methods

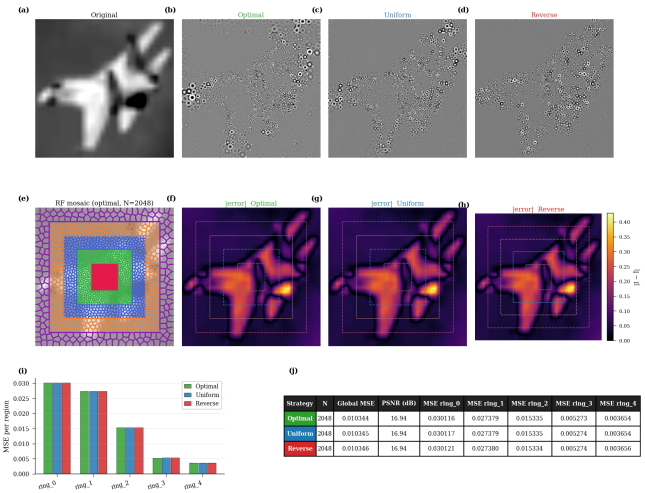
Our computational pipeline treats RGCs as mosaics of Difference-of-Gaussians (DoG) receptive fields tiled via k-means Voronoi cells, where each cell's center radius scales as $\sigma_c = s\sqrt{A_i}$ with its patch area A_i. Images are encoded into ON/OFF half-wave-rectified ganglion-cell responses and reconstructed via a minimum-norm matched-filter pseudoinverse decoder.

Allocation strategies are compared against rate-distortion theory (Ganguli & Simoncelli, 2010, 2014), which analytically predicts that optimal resources scale with $n_i \propto c_i^{1/(1+p)}$, where c_i is local gradient-energy complexity and p governs the distortion power law. We also test whether the optimal allocation is achievable under evolutionary algorithm schemes with random initializations.

Experiments use synthetic image ensembles with controlled regional statistics—horizontal bands of varying texture and concentric foveal rings—and compare the uniform, gradient-energy optimal, and adversarial reverse cell allocations.

Results

Across all tested budgets ($N = 8 \sim 8192$), the gradient-energy optimal allocation consistently yields the lowest reconstruction MSE, with the benefit growing monotonically with both budget size and inter-regional complexity contrast (Figures 1, 2). The reverse baseline consistently under-performs, confirming that allocation order matters.



Strategy	N	Global MSE	PSNR (dB)	MSE ring_0	MSE ring_1	MSE ring_2	MSE ring_3	MSE ring_4
Optimal	2048	0.010344	16.94	0.030116	0.027379	0.015335	0.005273	0.003654
Uniform	2048	0.010345	16.94	0.030117	0.027379	0.015335	0.005274	0.003654
Reverse	2048	0.010346	16.94	0.030121	0.027380	0.015334	0.005274	0.003656

Figure 1: **Retinal ganglion cell allocation strategies produce near-identical global reconstruction error despite differing spatial distributions.** (a) Original grayscale test image. (b–d) Reconstructed images under Optimal (gradient-energy), Uniform, and Reverse RGC allocations ($N = 2048$), with global MSE and PSNR shown above each. (e) Voronoi RF mosaic for the Optimal strategy, colored by concentric retinal ring. (f–h) Pixelwise absolute reconstruction error maps for each strategy; dashed boxes indicate ring boundaries. (i) Per-ring MSE broken down by retinal region, showing comparable errors across strategies at this budget. (j) Summary table of global and per-ring MSE and PSNR across all three strategies. Despite equivalent global MSE, the spatial structure of errors differs across allocation schemes, consistent with rate-distortion predictions.

Evolutionary learning converges in the direction of the theoretically optimal allocation and ultimately surpasses it, achieving lower MSE than the analytic solution (Figure 3).

Discussion

Taken together, these results support the normative hypothesis that the primate fovea's disproportionate ganglion-cell density is consistent with complexity-matched, rate-distortion-optimal coding.

In submission to the *9th Conference on Cognitive Computational Neuroscience* (CCN 2026).

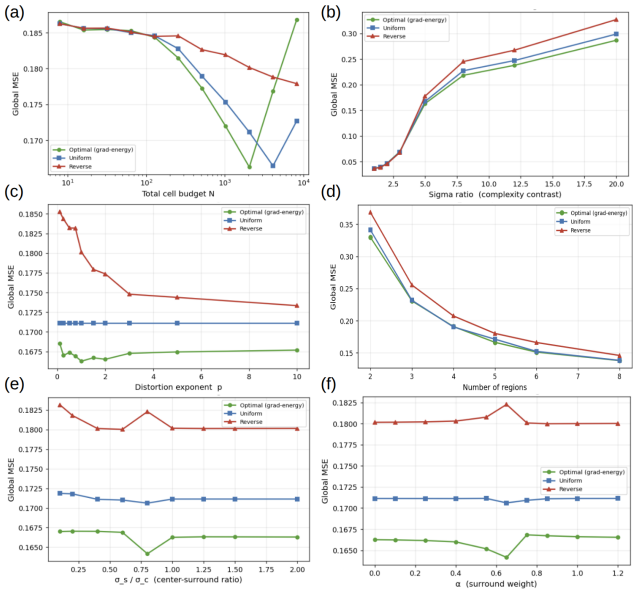


Figure 2: **Gradient-energy optimal allocation robustly outperforms uniform and reverse baselines across a wide range of model parameters.** Global MSE for Optimal (green), Uniform (blue), and Reverse (red) strategies as a function of: (a) total cell budget N, showing that the advantage of the optimal allocation grows with budget size; (b) complexity contrast (sigma ratio between regions), showing increasing benefit at higher inter-regional contrast; (c) distortion exponent p, where the optimal allocation is robust across a broad range of values; (d) number of retinal regions, confirming that results generalize beyond the five-ring default; (e) center-surround ratio σ_s/σ_c, illustrating robustness to DoG receptive field shape; and (f) surround weight α, showing consistent advantage across inhibitory surround strengths. In all panels, the Reverse allocation consistently yields the highest MSE, confirming that allocation order matters.

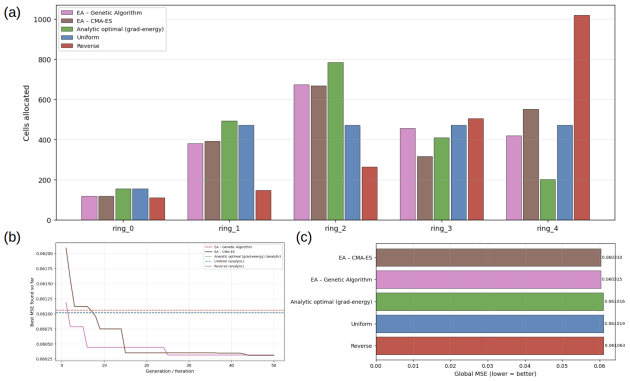


Figure 3: **Evolutionary algorithms independently recover near-optimal RGC allocations, surpassing the analytic gradient-energy solution.** (a) Per-ring cell counts for each strategy (N = 2048): Genetic Algorithm (pink), CMA-ES (brown), Analytic optimal (green), Uniform (blue), and Reverse (red). Both evolutionary methods allocate cells in a pattern broadly consistent with the analytic optimum, concentrating resources in mid-peripheral rings of higher complexity. (b) Convergence curves showing best MSE found so far as a function of generation/iteration; both EA methods descend below the analytic optimum (dashed green line) within 50 generations. (c) Final global MSE comparison across all strategies; EA methods achieve the lowest MSE, followed closely by the analytic optimum, with Uniform and Reverse allocations performing worse.

References

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